



JAIPUR NATIONAL UNIVERSITY

Semester/Year End/Back Examination: December-2023

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. CSE(Hons.) in Artificial Intelligence / B.Tech (Biomedical) / B.Tech (Chemical) / B.Tech (Civil Engineering) / B.Tech (Computer Science) / B.Tech (Cyber Security) / B.Tech (Data Science) / B.Tech (Electrical Engineering) / B.Tech (Mechanical)
Semester/Year:	I
Paper Code:	BTEMACO104T22
Title of Paper:	Engineering Mathematics I
Time: 3 Hours	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This paper consists of two sections, Section A and Section B. Section A consist of seven questions and Section B consists of six questions. Attempt any five questions from section A and four from section B.

SECTION-A

Marks
Allotted

Attempt any five questions.

Q.1 Find the asymptotes of the following curve $x^2y^2 - a^2(x^2 + y^2) = 0$.

4

Q.2 Define:

4

iii. Concavity and convexity

iv. Node, cusp and conjugate point.

- Q.3 Trace the curve $y^2(a+x) = x^2(a-x)$. 4
- Q.4 Find the second order Taylor's polynomial approximation of $f(x, y) = \sqrt{xy}$ about $(1, 3)$. 4
- Q.5 Prove that: - $B(m, n) = B(n, m)$. 4
- Q.6 Write the necessary condition of Lagrange's function for extreme points. 4
- Q.7 Write the statement of Green's Theorem and Gauss Divergence theorem 4

SECTION-B

Attempt any four questions.

- Q1 (a). Find the asymptotes of the curve $x^3 + 2x^2y - xy^2 - 2y^3 + 3xy + 3y^2 + x + 1 = 0$. (5)
- (b). Show that the points of inflexion of the curve $y^2 = (x-a)^2(x-b)$ lie on the straight line. (5)
 $3x+a=4b$.
- Q2. State and prove the Euler's theorem. Also, if $u = x \sin^{-1} \left(\frac{y}{x} \right)$ then prove that (10)

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0$$
- Q3. Find the dimensions of the rectangular box open at the top of maximum capacity whose surface is 108 sq.cm (10)
- Q4. Find the minimum value of $u = x^2 + y^2 + z^2$ when $x + y + z = 3a$. (10)
- Q5 (a). Evaluate:
 (a) $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dy dx$. (5)
- (b) Find the area enclosed by the parabolas $y^2 = 4ax$ and $x^2 = 4ay$. (5)
- Q6. Use Stokes' Theorem to evaluate $\int_C \vec{F} \cdot d\vec{r}$, where $F = (3yx^2 + z^3)i + y^2j + 4yx^2k$ C is triangle with vertices $(0, 0, 3)$, $(0, 2, 0)$ and $(4, 0, 0)$. C has a counter clockwise rotation if you are above the triangle and looking down towards the xy -plane. (10)